

Chapter 3

Chemical Bonding and Structure

Learning Objectives

- Describe the formation of covalent bond, ionic bond and metallic bond.
- Understand Lewis structure.
- Explain the terms dipole and dipole-moment, polar and non-polar.
- Predict polarity of bonds and molecules using electronegativity and dipole moment.
- Explain bond length, bond angle, bond energy and bond order.
- Predict the molecular geometry by VSEPR theory.
- Predict the hybridization of atoms in a molecule.
- Relate the physical properties such as boiling points and solubility with types of intermolecular forces.
- Explain the periodic trend of lattice energy.

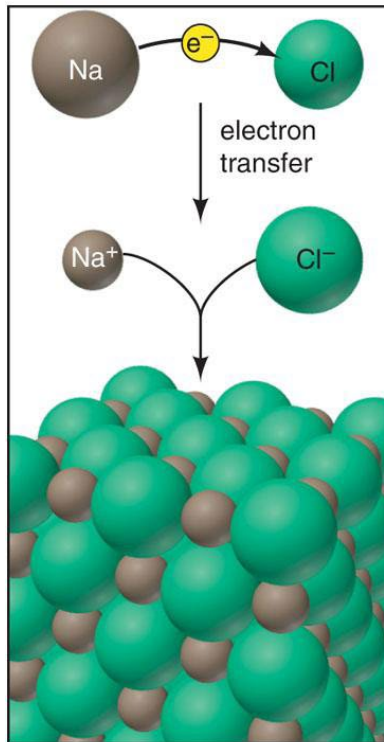
Why do atoms form bond?

- In any compound, atoms are “bonded”, by covalent / ionic / metallic bond.
- Why? Because the system (atoms) becomes overall more stable.
- When will covalent bond form?
- When will ionic bond form?

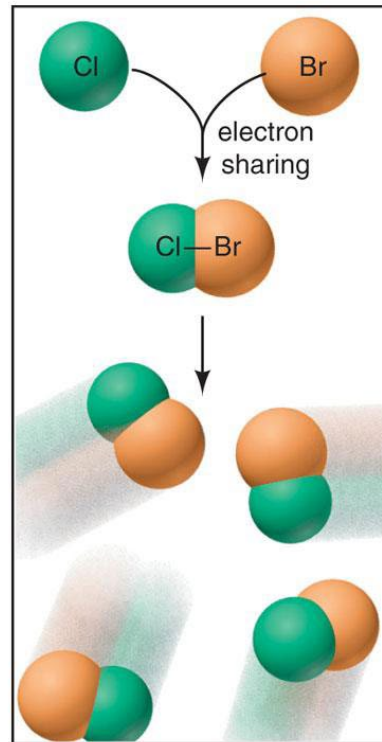
H 2.20		Electronegativity Chart																He	
		Using the Pauling scale																	
Li 0.98	Be 1.57											B 2.04	C 2.55	N 3.04	O 3.44	F 3.98	Ne		
Na 0.93	Mg 1.31											Al 1.61	Si 1.90	P 2.19	S 2.58	Cl 3.16	Ar		
K 0.82	Ca 1.00	Sc 1.36	Ti 1.54	V 1.63	Cr 1.66	Mn 1.55	Fe 1.83	Co 1.88	Ni 1.91	Cu 1.90	Zn 1.65	Ga 1.81	Ge 2.01	As 2.18	Se 2.55	Br 2.96	Kr 3.00		
Rb 0.82	Sr 0.95	Y 1.22	Zr 1.33	Nb 1.6	Mo 2.16	Tc 1.9	Ru 2.2	Rh 2.28	Pd 2.20	Ag 1.93	Cd 1.69	In 1.78	Sn 1.96	Sb 2.05	Te 2.1	I 2.66	Xe 2.60		
Cs 0.79	Ba 0.89	Lu 1.27	Hf 1.3	Ta 1.5	W 2.36	Re 1.9	Os 2.2	Ir 2.20	Pt 2.28	Au 2.54	Hg 2.00	Tl 1.62	Pb 2.33	Bi 2.02	Po 2.0	At 2.2	Rn 2.2		
Fr >0.79	Ra 0.9	Lr 1.3	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv ³	Ts	Og		

Three types of bonds

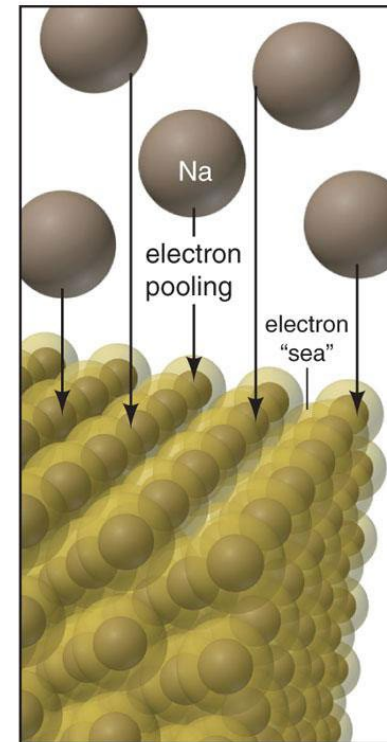
Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



A Ionic bonding



B Covalent bonding



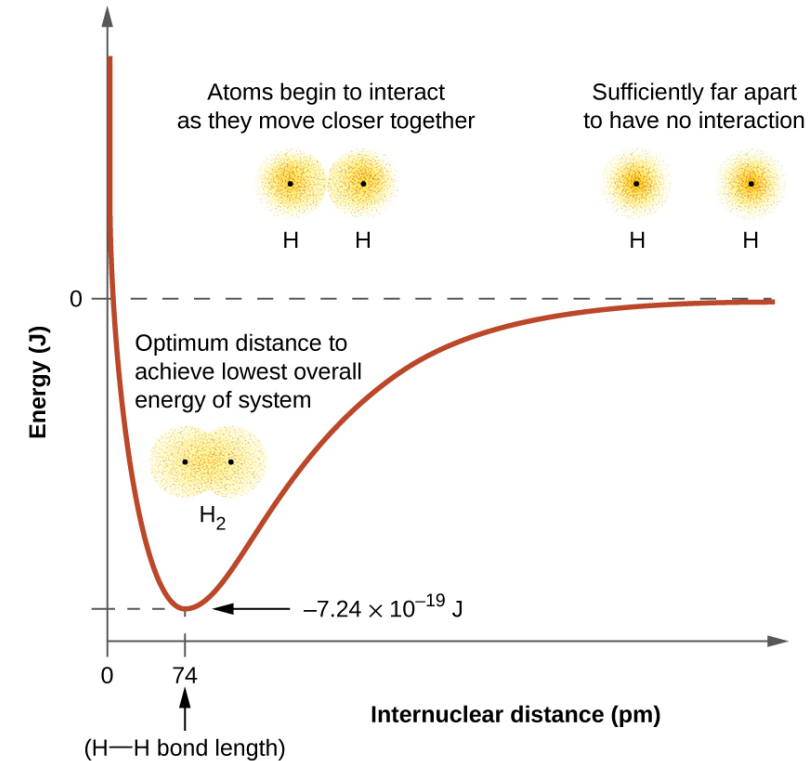
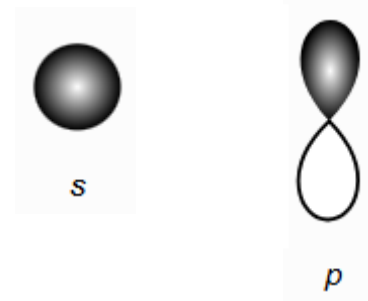
C Metallic bonding

Part 1

Covalent Bond

How are covalent bonds form?

- Valence bond theory
 - Overlap of orbitals (s and p)
 - Orbital must be in-phase
 - Different ways of overlap: σ and π
 - Spin of electrons must pair up
- Which overlap forms a bond?
- Which overlap does not form a bond?



Lewis Structure

- Dot-and-cross structure

- Bonding pair (bp): a line or two dots
- Lone pair (lp): two dots
- Single bond / double bond / triple bond and bond length

Bond length

Bond angle

Bond energy

- Octet rule

- Exception: H
- Exception: B (and Be)
- Exception: hypervalent
- Dative bond (coordinate bond)
- Formal charge

Bond order

Lewis Structure: exercise

- N_2 , H_2O , NH_3 , NH_4^+ , CO_2
- PCl_3 , PCl_5 , SF_6 , CO
- CH_4 , HCN , C_2H_4
- H_2SO_4 , H_3PO_4 , HNO_3

Resonance

- How to explain the bond length of:



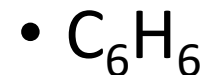
Resonance structure



Resonance hybrid



Delocalization



Double-headed arrow

Bond Energy and Bond Length

- Bond energy: the energy (standard enthalpy) change for breaking the bond in 1 mol of gaseous molecules. It is also called bond strength or bond enthalpy.



- Bond energy is always positive.
- The bond length of a covalent bond is the distance between the nuclei of two bonded atoms.
- Generally, the longer the bond length, its bond energy is _____

Average bond energies and bond lengths

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

Table 9.2 Average Bond Energies (kJ/mol) and Bond Lengths (pm)

Bond	Energy	Length	Bond	Energy	Length	Bond	Energy	Length	Bond	Energy	Length
Single Bonds											
H—H	432	74	N—H	391	101	Si—H	323	148	S—H	347	134
H—F	565	92	N—N	160	146	Si—Si	226	234	S—S	266	204
H—Cl	427	127	N—P	209	177	Si—O	368	161	S—F	327	158
H—Br	363	141	N—O	201	144	Si—S	226	210	S—Cl	271	201
H—I	295	161	N—F	272	139	Si—F	565	156	S—Br	218	225
			N—Cl	200	191	Si—Cl	381	204	S—I	~170	234
C—H	413	109	N—Br	243	214	Si—Br	310	216			
C—C	347	154	N—I	159	222	Si—I	234	240	F—F	159	143
C—Si	301	186							F—Cl	193	166
C—N	305	147	O—H	467	96	P—H	320	142	F—Br	212	178
C—O	358	143	O—P	351	160	P—Si	213	227	F—I	263	187
C—P	264	187	O—O	204	148	P—P	200	221	Cl—Cl	243	199
C—S	259	181	O—S	265	151	P—F	490	156	Cl—Br	215	214
C—F	453	133	O—F	190	142	P—Cl	331	204	Cl—I	208	243
C—Cl	339	177	O—Cl	203	164	P—Br	272	222	Br—Br	193	228
C—Br	276	194	O—Br	234	172	P—I	184	243	Br—I	175	248
C—I	216	213	O—I	234	194				I—I	151	266
Multiple Bonds											
C=C	614	134	N=N	418	122	C C	839	121	N N	945	110
C=N	615	127	N=O	607	120	C N	891	115	N O	631	106
C=O	745	123	O ₂	498	121	C O	1070	113			
(799 in CO ₂)											

Bond lengths and bond energies vary slightly from molecule to molecule. Hence average values are tabulated.

The Relation of Bond Order, Bond Length and Bond Energy

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

Table 9.3 The Relation of Bond Order, Bond Length, and Bond Energy

Bond	Bond Order	Average Bond Length (pm)	Average Bond Energy (kJ/mol)
C—O	1	143	358
C=O	2	123	745
C≡O	3	113	1070
C—C	1	154	347
C=C	2	134	614
C≡C	3	121	839
N—N	1	146	160
N=N	2	122	418
N≡N	3	110	945

Why do covalent substances melt and boil at low temperatures?

The two forces in covalent molecules are:

1. Strong covalent bonds that hold the atoms together within the molecule
2. Weak intermolecular forces (IMF) that act between the molecules.

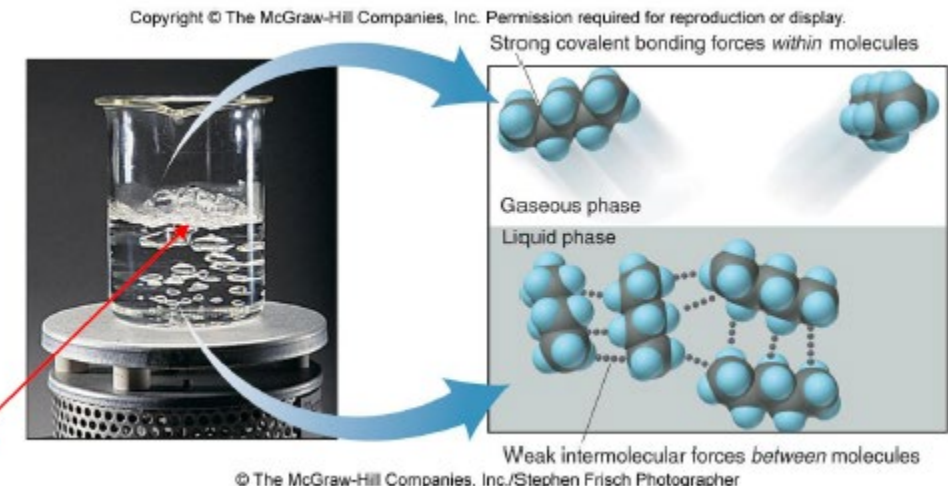
When the covalent liquid boils, it is the weak IMF between the molecules that are being overcome and not the strong covalent bonds → less energy needed → lower boiling point

BUT

This reasoning only applies to simple molecular substances.

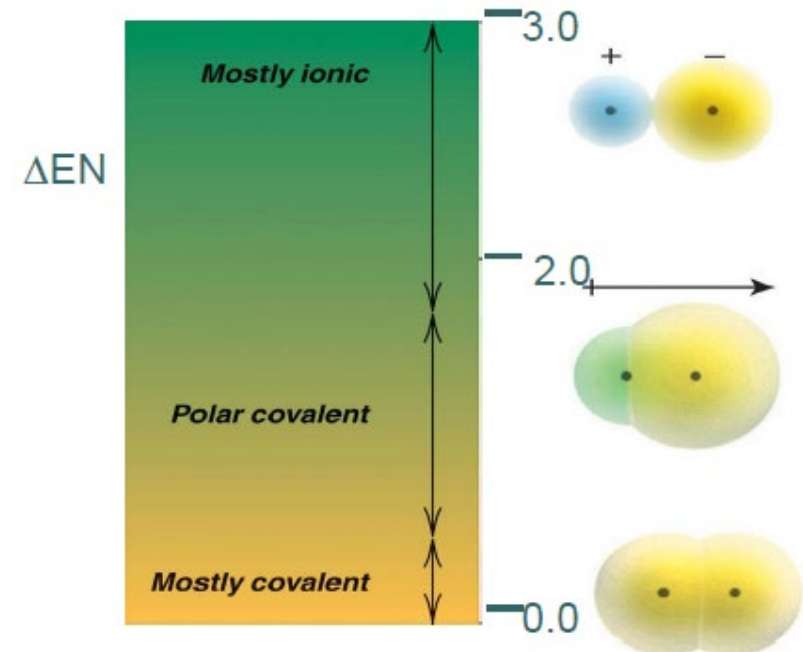
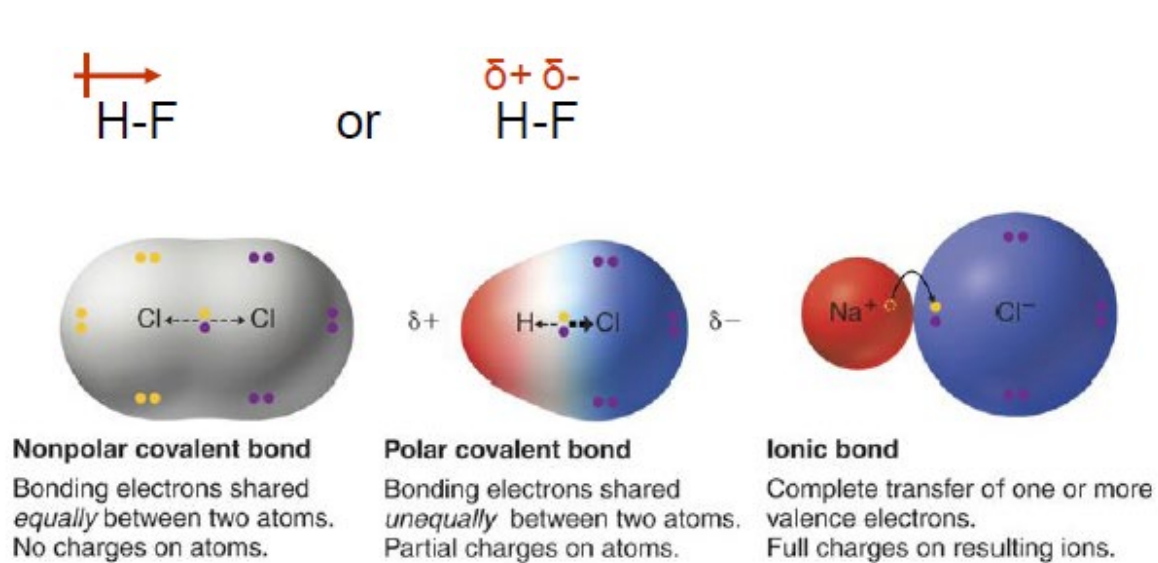
For giant covalent substances (diamond, graphite, silicon, SiO_2 , ...): need to break (a lot of) covalent bonds when melting → melting point is high

Pentane



Polar Covalent Bonds and Bond Polarity

- When the bonded atoms are of different electronegativities, the bonding pair is shared unequally. This leads to partially negative and positive poles in the bond. Such a polar covalent bond is depicted by a polar arrow $\rightarrow$ pointing towards the negative pole.
- The polarity of the entire molecule depends on the bond polarity and the geometry of the molecule [vector addition of dipoles].



VSEPR and Molecular Geometry

- How to deduce the geometry of the molecule?
- Valence Shell Electron Pair Repulsion Theory
 - Repulsions between electrons in **bonds** and **lone pairs** determines the overall shape of a molecule
 - Inner electrons are not considered → valence shell
 - Electrons are typically in pairs [if there is a single electron, treat as a “pair”]
- Separate n pairs as far away as possible:

2

3

4

5

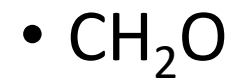
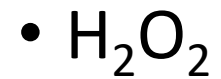
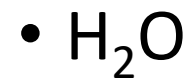
6

VSEPR – when electrons are “invisible”

Number of lone pair(s)

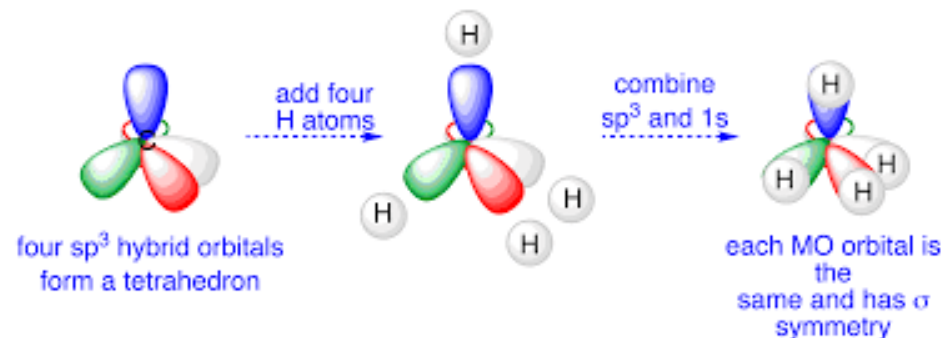
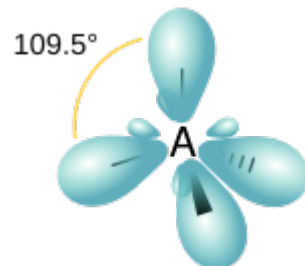
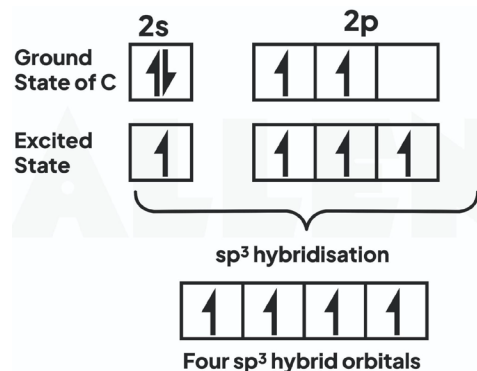
No. of electron domain (bp+lp)	0	1	2	3	4
					Lone pair at equatorial

VSEPR – exercise



Explanation of geometry: Hybridization

- Methane (CH_4) geometry:
- Electron configuration of C:
- What is the angle between the atomic orbitals?
- Pauling's idea of hybridization: the valence atomic orbitals in the molecule are different from those in the isolated atoms. The atomic orbitals can mix to obtain new orbitals. The process of mixing is called hybridization and the new orbitals are called hybrid orbitals.



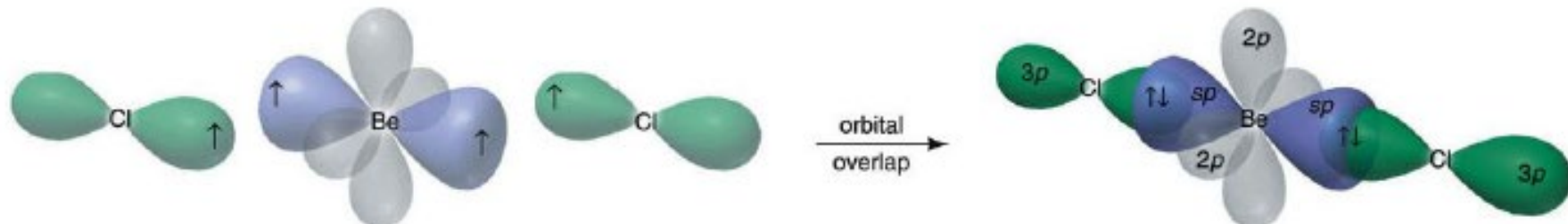
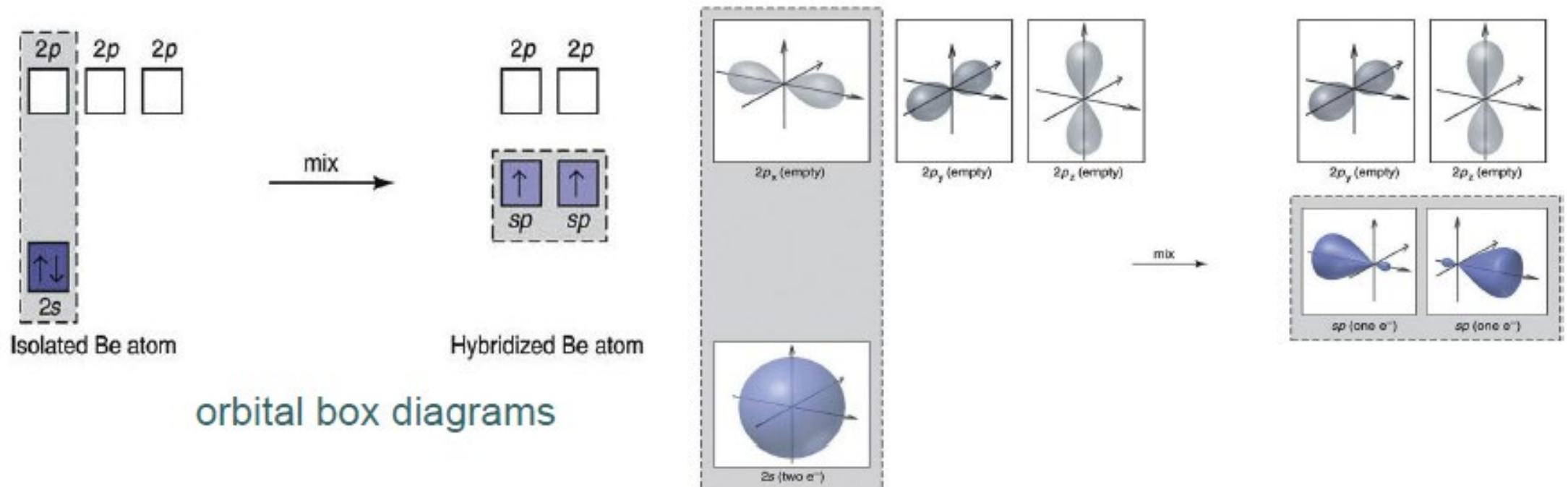
Hybridization of ethene and ethyne

- sp^3

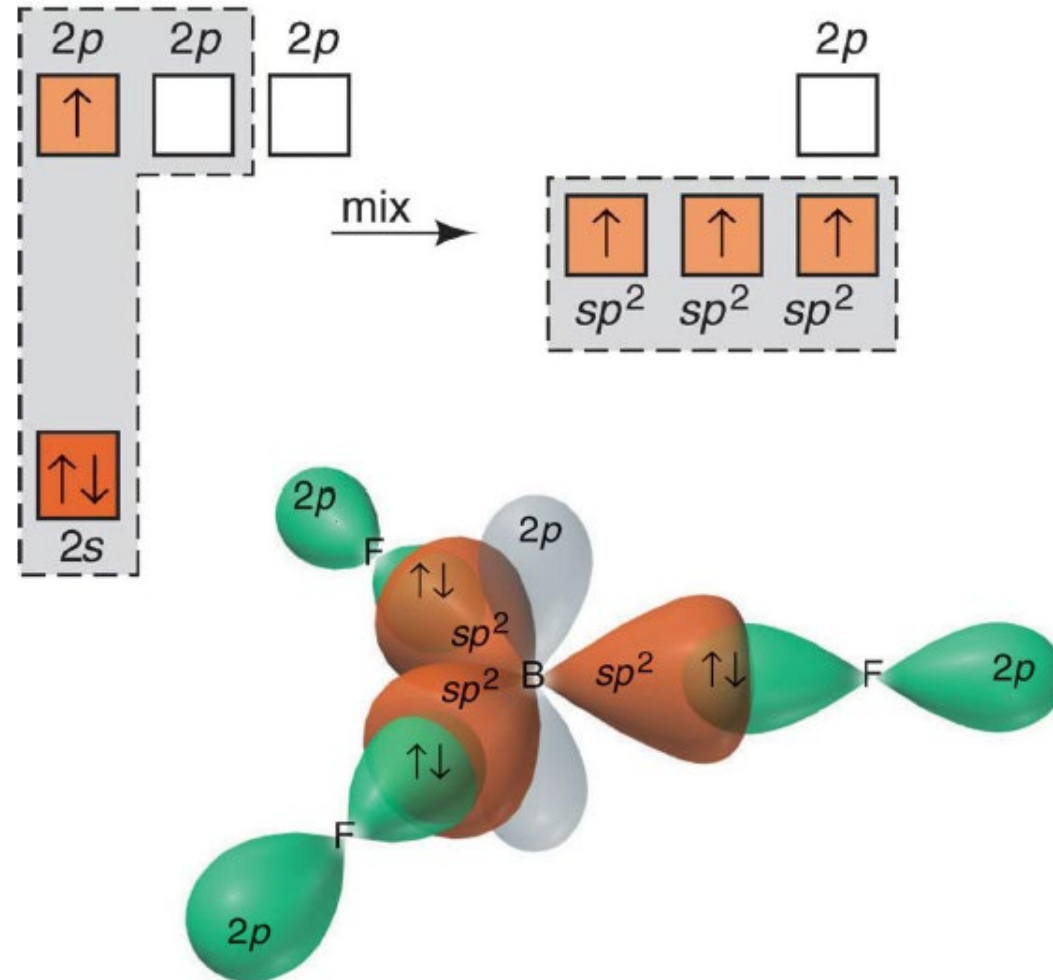
- sp^2

- sp

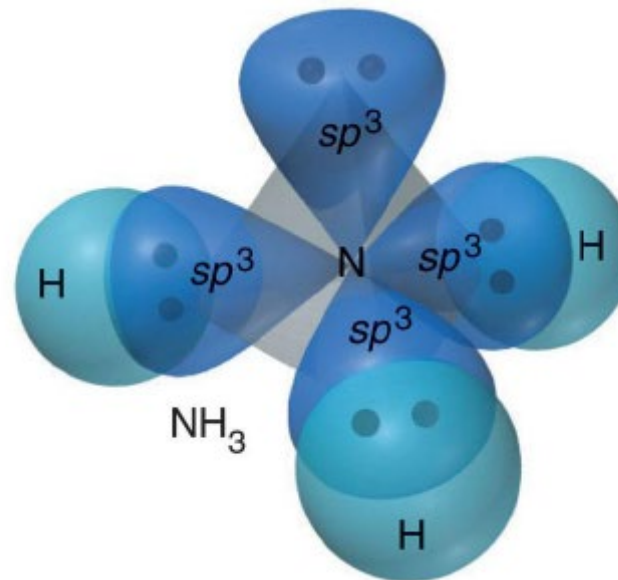
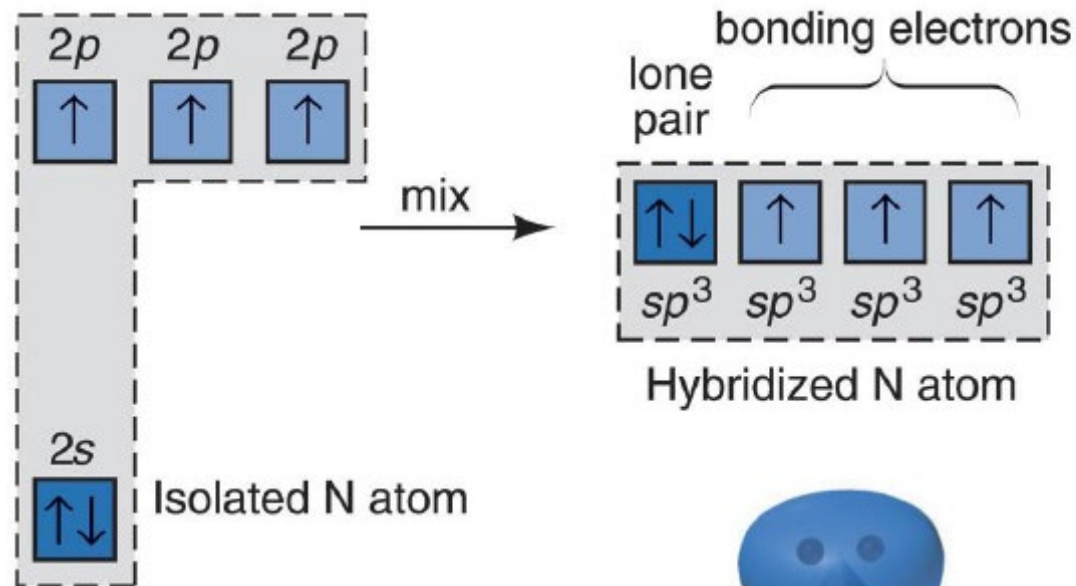
More examples of hybridization (self study)



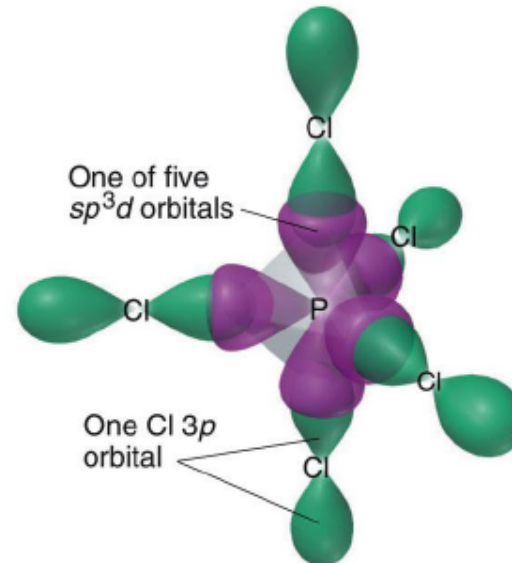
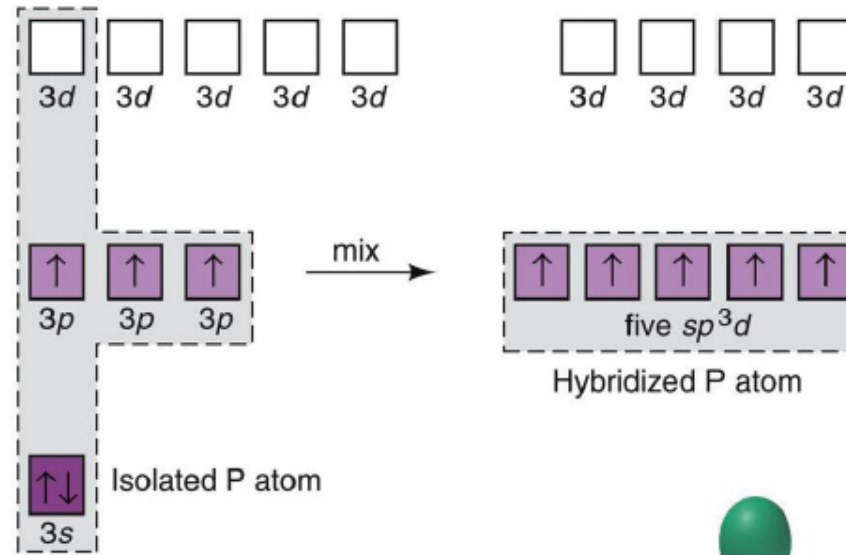
More examples of hybridization (self study)



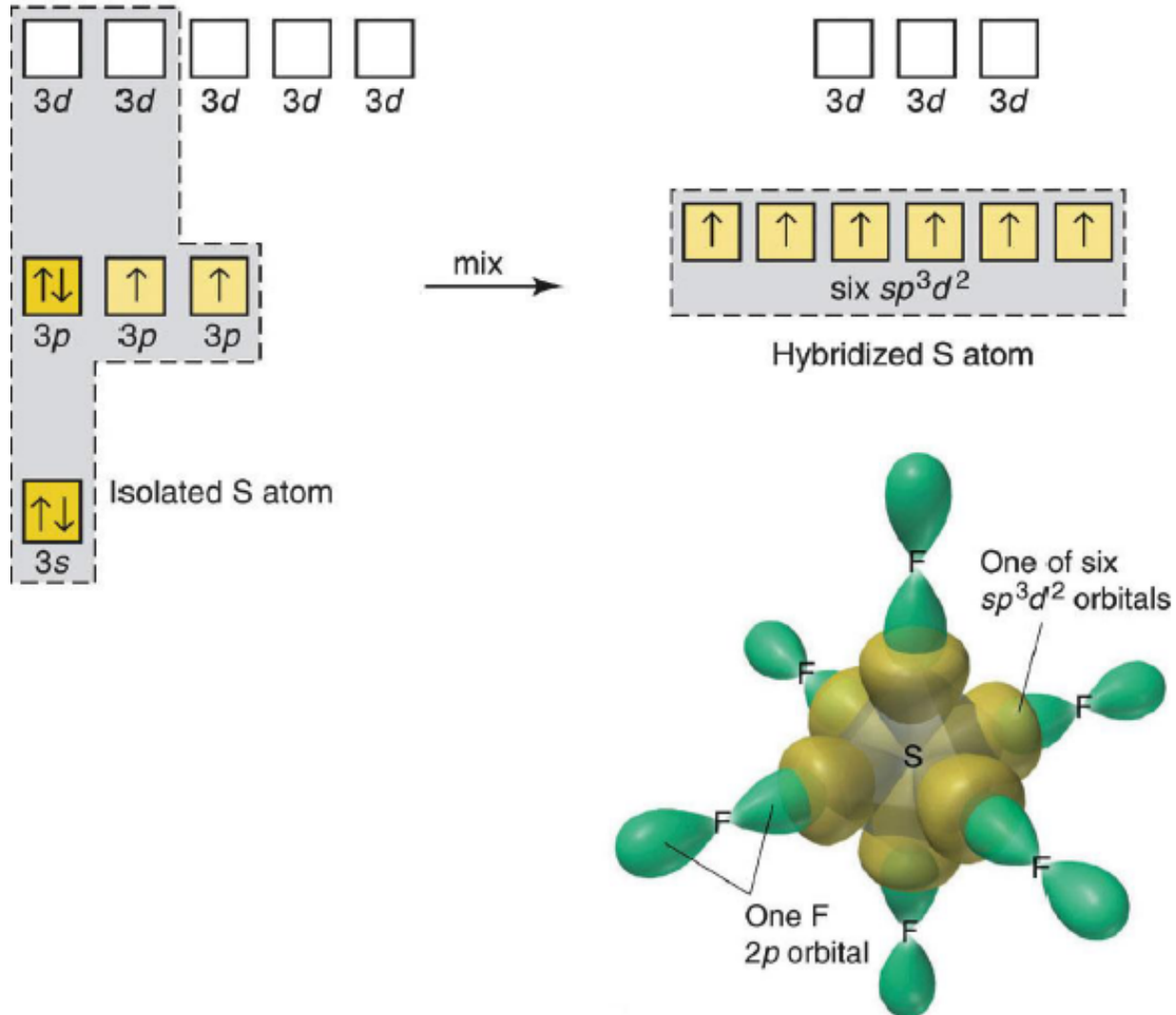
More examples of hybridization (self study)



More examples of hybridization (self study)



More examples of hybridization (self study)



More examples of hybridization (self study)

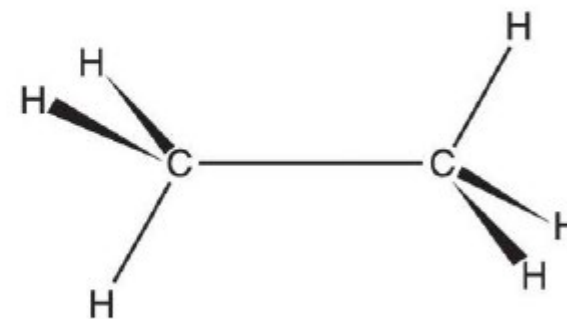
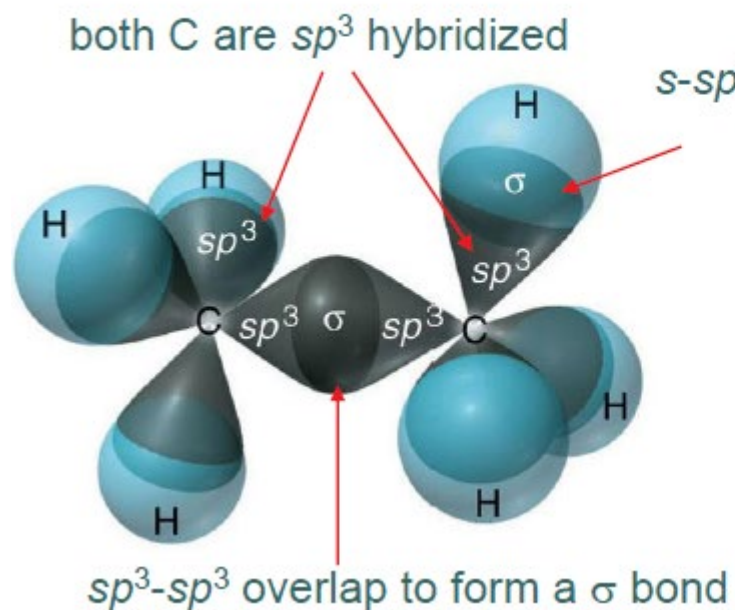
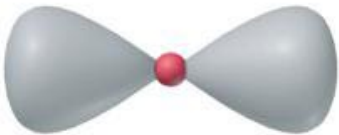
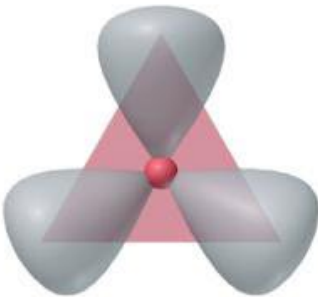
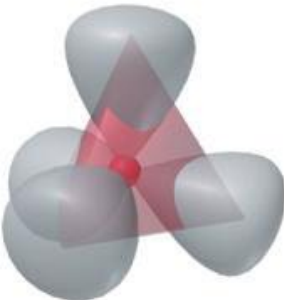
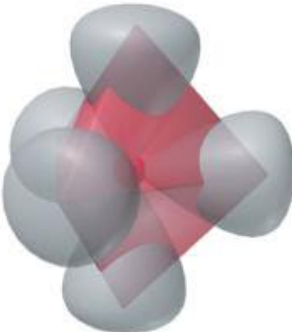
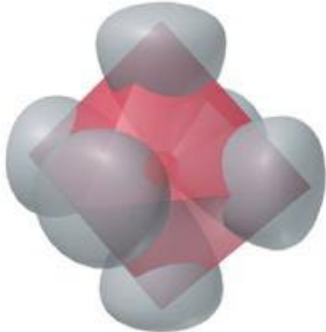


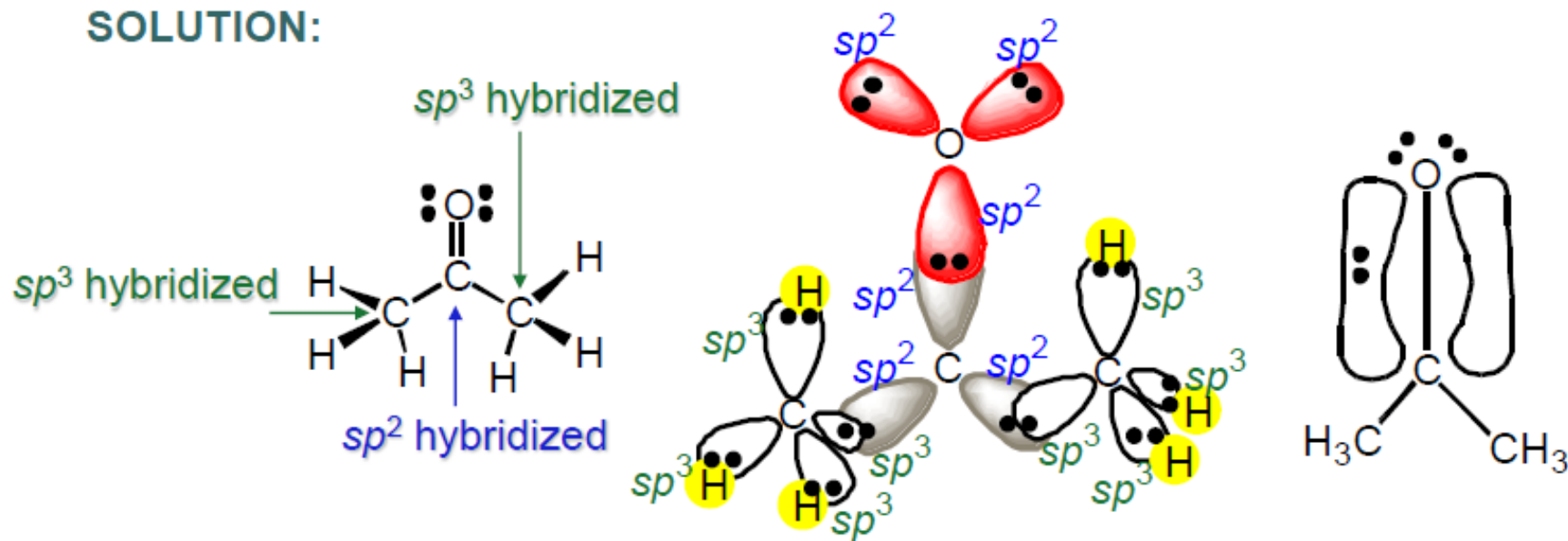
Table 11.1 Composition and Orientation of Hybrid Orbitals

	Linear	Trigonal Planar	Tetrahedral	Trigonal Bipyramidal	Octahedral
Atomic orbitals mixed	one s one p	one s two p	one s three p	one s three p one d	one s three p two d
Hybrid orbitals formed	two sp	three sp^2	four sp^3	five sp^3d	six sp^3d^2
Unhybridized orbitals remaining	two p	one p	none	four d	three d
Orientation					

Exercise

PROBLEM: Describe the types of bonds and orbitals in acetone, $(\text{CH}_3)_2\text{CO}$.

SOLUTION:



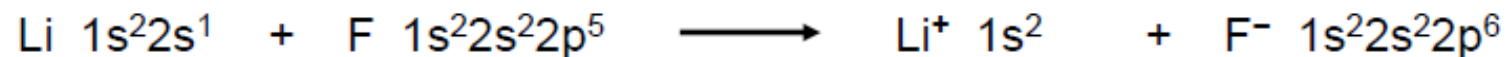
Practice question Describe the types of bonds and orbitals in (a) HCN (b) CO_2

Part 2

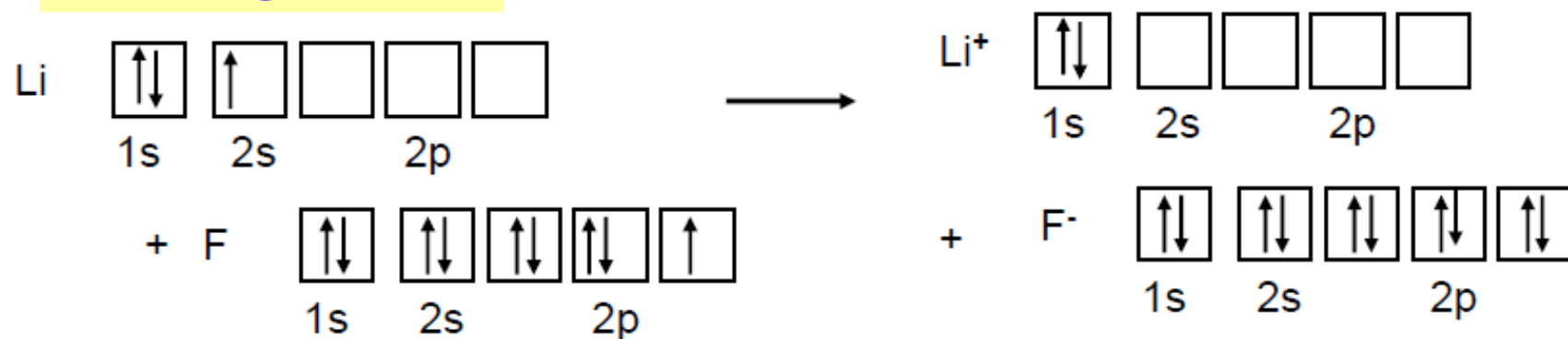
Ionic Bond and Metallic Bond

Ionic Bonding Model

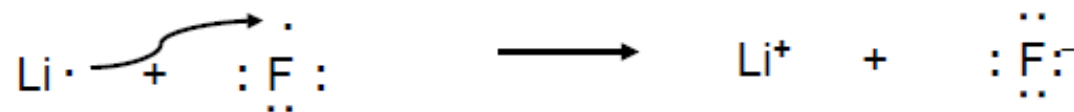
Electron configurations



Orbital diagrams

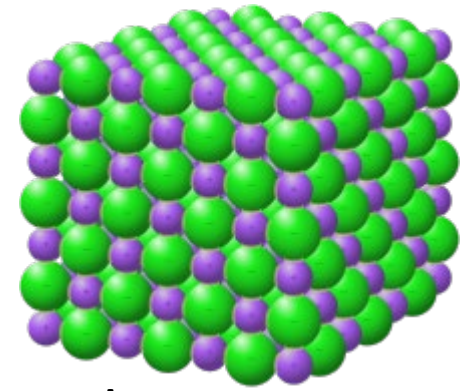


Lewis electron-dot symbols



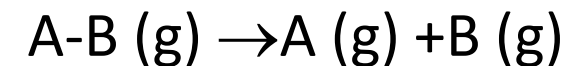
Problem: Depict the formation of Potassium Oxide and Barium chloride

Ionic Lattice



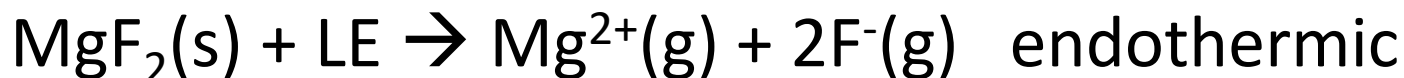
- Cations and anions (infinite number of each) are held together by electrostatic forces [ionic bond] to form a regular, repeating, 3D giant structure, called lattice.
- In the solid form, the ions are “fixed” at its own position and cannot move around. They can at most vibrate a little bit around its equilibrium position. That’s why solid salt is not a conductor.
- In the molten state, the ions can freely move, therefore it becomes a conductor of electricity, hence electrolysis becomes possible.
- When ionic compound dissolves in solvent (water), the ions can also freely move → conductor.

Lattice Energy



$$\Delta H^{\circ}_{\text{bond breaking}} = BE_{A-B}$$

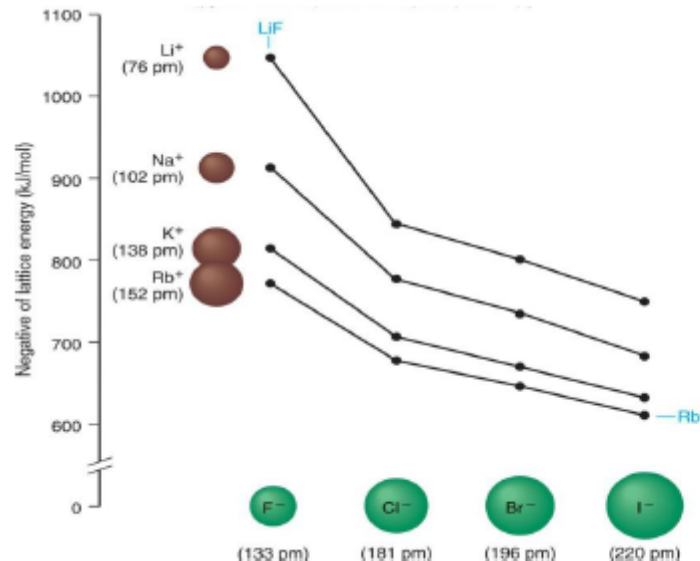
- Lattice energy is the energy released when **gaseous cations and anions** combine to form 1 mole of ionic compound (solid).
- It is equivalent to the energy needed to break all the ionic bonds in one mole of ionic compound, to form **gaseous cations and anions**.



- Lattice energy is the “bond energy” of ionic bond
 - Ionic bond is not localized between two ions, so we cannot just break one Mg-F bond, rather all the attractions between all Mg^{2+} and all F^{-} has to be broken.
- The larger the lattice energy, the stronger the ionic bond.

Periodic Trends in Lattice Energy

- Lattice Energy cannot be directly measured → Born-Haber cycle is used for calculation [we will cover this in Chapter 7]
- Lattice energy is proportional to the electric potential energy
- Lattice energy $\propto$ electric potential energy $\propto \frac{q_1 \times q_2}{r_1 + r_2}$
 - q_1 and q_2 are the charges of the ions; r_1 and r_2 are the radii of the ions



Increase in the radii of ions leads to decrease in the lattice energy.(refer Graph)

Lattice energy increases with the Increase in the ionic charge.

The lattice energy of LiF and MgO are 1050 and 3923 kJ/mol. why the ratio is 1:4?

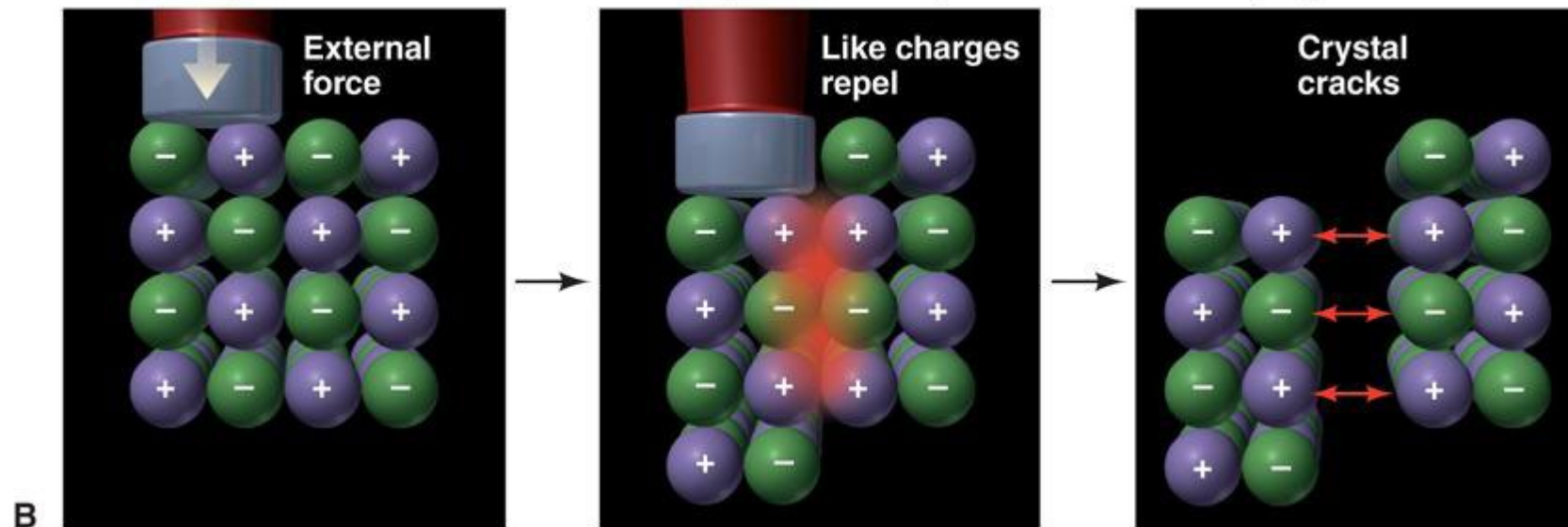
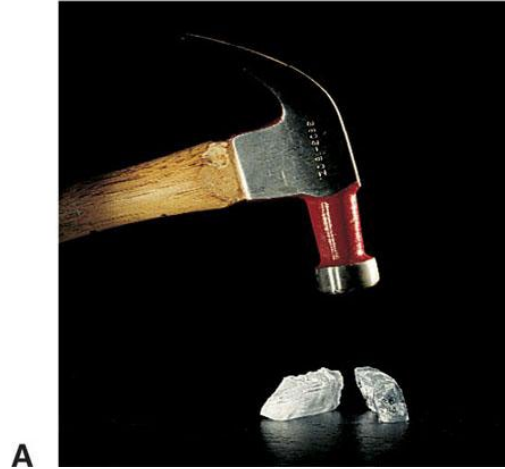
Guess the approximate ratio of NaF and AlPO_4

Properties of Ionic Compounds

- A typical ionic compound is hard and brittle.
 - This is because there are strong attractive forces that hold the ions together.
- Ionic compounds usually have high melting and boiling points.
 - For melting/boiling, ionic bonds need to be broken, requiring large amount of energy.
- Ionic compounds do not conduct electricity in the solid state but do so when melted or dissolved.
 - This is because the ions can move in the molten or dissolved state

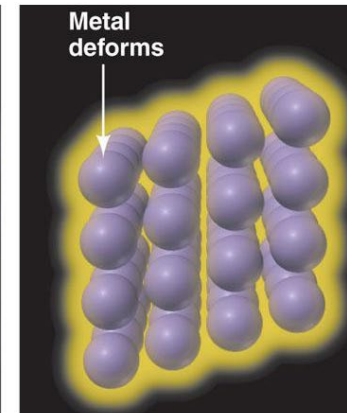
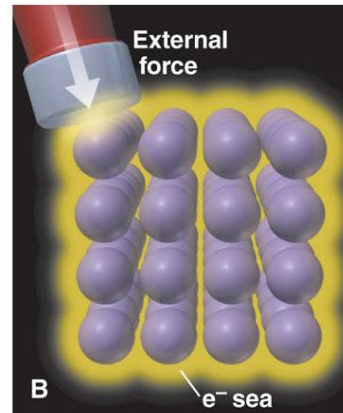
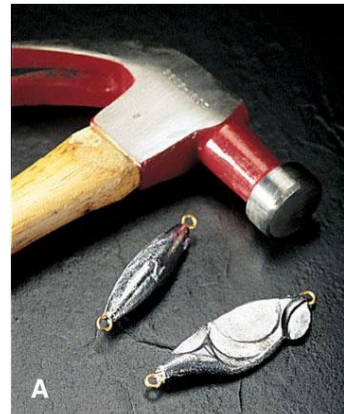
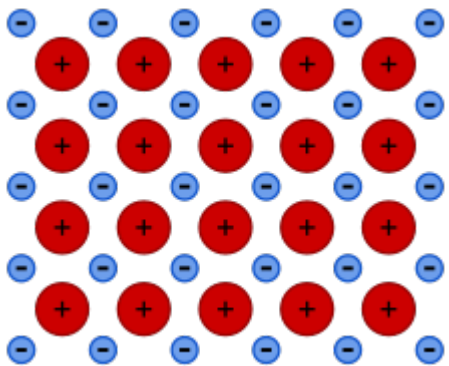
Compound	mp ($^{\circ}\text{C}$)	bp ($^{\circ}\text{C}$)
CsBr	636	1300
NaI	661	1304
MgCl ₂	714	1412
KBr	734	1435
CaCl ₂	782	>1600
NaCl	801	1413
LiF	845	1676
KF	858	1505
MgO	2852	3600

Why do ionic solids crack?



Metallic Bond Model

- All the atoms in a piece of metal contribute their valence electrons to form a delocalized free electron “sea”.
- The piece is held together by the mutual attraction of the metal cations and the mobile electrons.
- Since electrons can freely move, the metal is a conductor.
- The metal ion array is regular but not rigid → ductile and malleable. 延展性



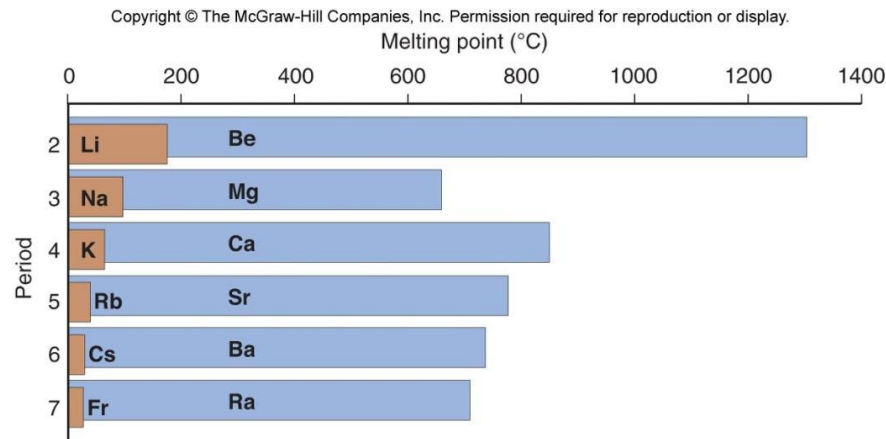
Metal ions slide past each other through the electron sea and end up in new positions.

Properties of metals

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

Table 9.6 Melting and Boiling Points of Some Metals

Element	mp (°C)	bp (°C)
Lithium (Li)	180	1347
Tin (Sn)	232	2623
Aluminum (Al)	660	2467
Barium (Ba)	727	1850
Silver (Ag)	961	2155
Copper (Cu)	1083	2570
Uranium (U)	1130	3930



- Melting points are only moderately high.
- **Reason:** the cations can move without breaking the attraction to the surrounding electrons
- Boiling points are very high
- **Reason:** Higher energy is needed to break the cation from all the valence electrons.
- Periodic trends: m.p. decreases down the group.
- **Reason:** Larger metal ions have a weaker attraction to the electron sea.
- m.p. increases across the period.
- **Reason:** The charge on the cation increases from left to right, hence stronger attraction towards the electron sea.